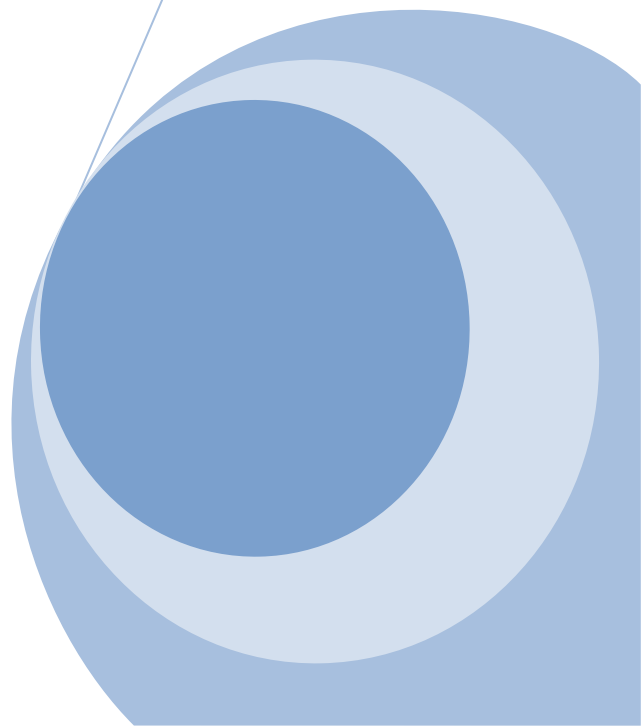




PART-1

INTRODUCTION OF JAVA

BY
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Introduction

- ❖ Java is High level, Simple, Platform Independent, Architecture-Neutral, Secure, Robust, Multithreaded, Distributed, Object Oriented programming language.
- ❖ It is intended to let application developers "write once, run anywhere" (WORA), meaning that code that runs on one platform does not need to be recompiled to run on another. Java applications are typically compiled to byte code (class file) that can run on any Java virtual machine (JVM) regardless of computer architecture .
- ❖ Java is one of the most popular programming languages in use, particularly for client-server web applications.

Features/Buzzwords: -**➡ Simple:-**

- Java was designed to make it much easier to write bug free code.
- There aren't a lot of special cases or tricks that will confuse beginners.
- The runtime environment provides automatic memory allocation and garbage collection so there's less for the programmer to think about.

➡ Secure:-

- Java program run within the JVM which protected from unauthorized or illegal access to system resources.

➡ Platform Independent: -

- When you compile Java program on one Operating System (OS) then compiler generate Byte Code. You can run that Byte code on any OS which has the JVM.

➡ Architecture-neutral: -

- The byte code is independent of the underlying platform that the program is running on. For example, it doesn't matter if your operating system is 32-bit or 64-bit; the Java byte code is exactly the same. You don't have to recompile your Java source code for 32-bit or 64-bit. (So, "architecture" refers to the CPU architecture).

➡ Robust: -

- Java is robust because it having Strong memory management, No Pointers, Exception handling, Platform Independent.

➡ Multithreaded: -

- It supports multithreaded programming, which allows us to write programs that do many things simultaneously.

➡ Distributed:-

- Using RMI and EJB we can create distributed application in java.

➡ Object Oriented:-

- It helps to break the complex code into easy to understand objects.

Programming Language

Program: - Set of Instructions for a device to perform some specific task.

Language: - It is the medium of communication.

A **programming language** is a formal constructed language designed to communicate instructions to a machine, particularly a computer. Programming languages can be used to create programs to control the behaviour of a machine or to express algorithms.

➡ **Type of Programming language:**

1. Low level/Machine level/Native code/Binary code
2. Assembly/Middle level/byte code
3. High level/Source code

1. Low level Programming language

- ⇒ Binary digits (0, 1) will be used to write the instruction.
- ⇒ It will be understood by the machine directly.
- ⇒ No translator program is required

Problem:

- ⇒ It is very complex to write, Modify and debug the instruction.
- ⇒ Binary code is platform dependent.

2. Middle level Programming language

- ⇒ Assembly code (like ADD, SUB, MUL, DIV etc) will be used to write the instruction.
- ⇒ Instruction written in assembly language will not be understood by the machine directly.
- ⇒ Assembler is use for convert assembly language to Machine language.

Problem:

- ⇒ The code written in Assembly language will be processor dependent.

3. High level Programming language

- ⇒ It is understand by Human, English words is use to write the instruction.
- ⇒ It is not understand by machine directly.
- ⇒ Compiler and Interpreter are used to convert High level Language to Machine Language.

Advantage:

- ⇒ It is easy to write, Modify and debug the instructions written using High Level Language.

Note:-

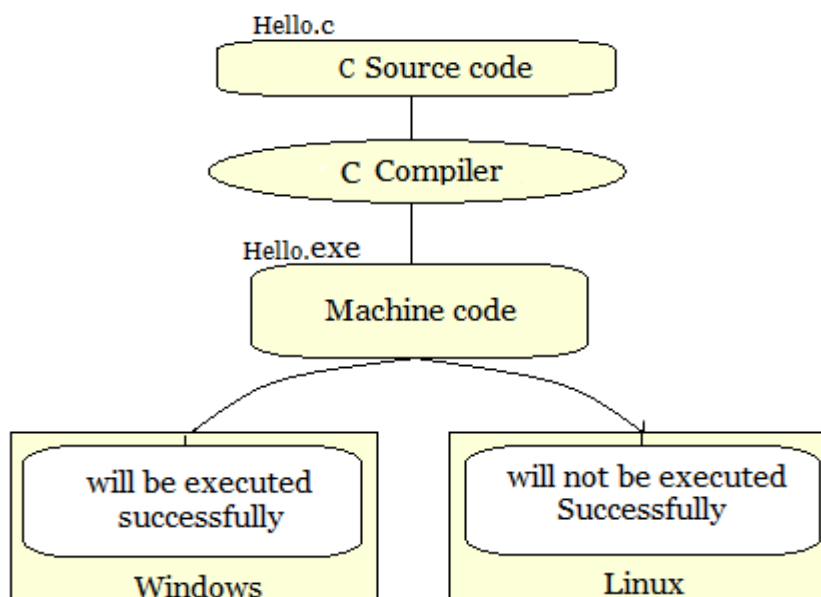
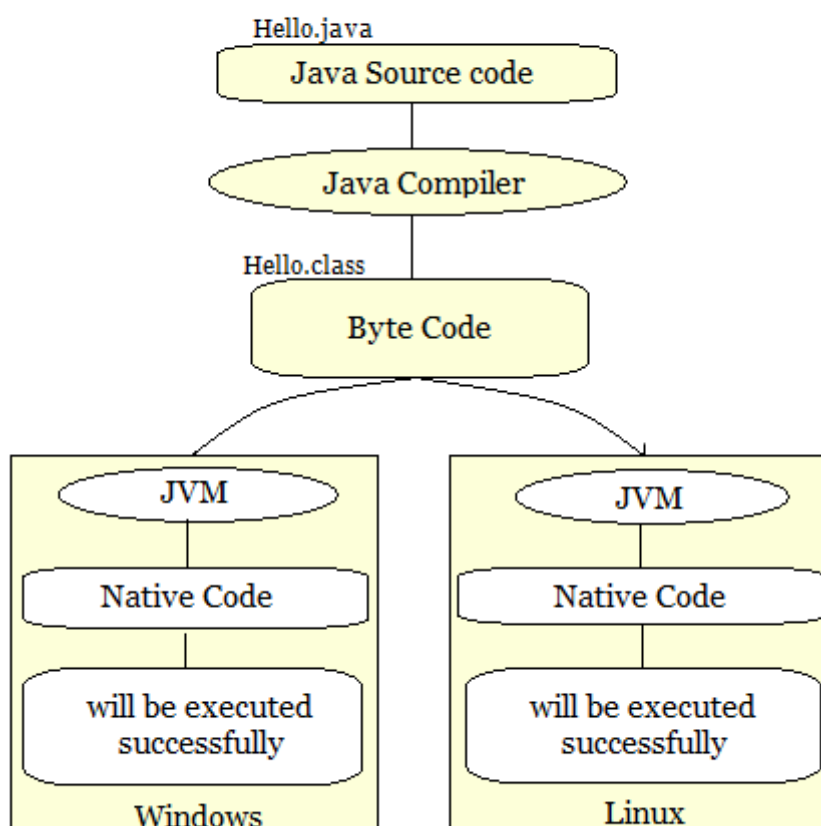
C /C++ program execution is faster than JAVA Program. Because

C/C++ compilation and execution step:-

Source code → Binary code

JAVA compilation and execution step:-

Source code → Byte Code → Binary code

C Program Development and Execution ModalJava Program Development and Execution Modal

Java Compiler:-

- ❖ Java Compiler is a program developed in C and C++ programming language with the name “**javac**”.
- ❖ It will check syntactical of grammatical error of the programs.
- ❖ It converts source code to byte code.
- ❖ Java Compiler (javac) is platform dependent.

Java Interpreter:-

- ❖ Java Interpreter is a program developed in C and C++ programming language with the name “**java**”.
- ❖ It converts byte code to machine code.
- ❖ It executes that native code.
- ❖ Java Interpreter is platform dependent.

JIT (Just-In-Time):-

- ❖ The Just-In-Time compilation (also known as dynamic translation or JIT) is a technique used in computing to improve the quality of the runtime performance of a computer program.
- ❖ JIT is capable of enforcing security guarantees.
- ❖ Just-in-time compilation is the conversion of non-native code, for example byte code, into native code just before it is executed.
- ❖ JIT is the component of the Java Runtime Environment (JRE).

JDK (Java Development Kit)/SDK (Software Development Kit):-

- ❖ JDK is a set of various utility programs which are required for developing and executing the java programs.
- ❖ JDK is Platform dependent.

Following are various utility programs provided under JDK.

1. Java Development Tools
 - I. javac
 - II. java
 - III. javap
 - IV. jar etc
2. Source Files
3. JRE

JRE (Java Runtime Environment):-

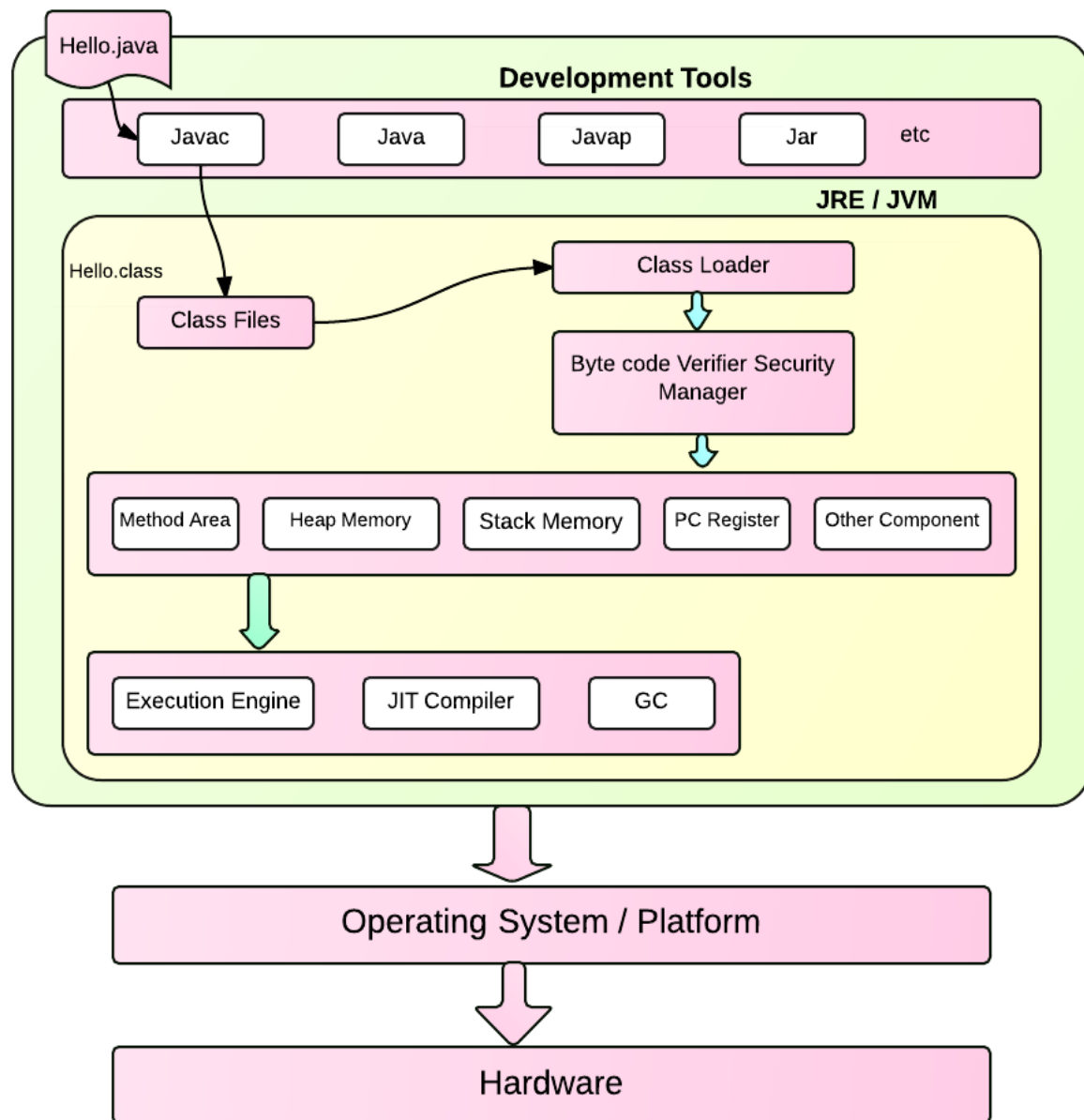
- ❖ JRE is implementation of JVM. It contains class libraries, Interpreter, JIT compiler etc.
- ❖ Only JRE is enough to run the Java Program.
- ❖ JRE is Platform independent.

JVM (Java Virtual Machine):-

- ❖ JVM implementation provides an environment of run our Java applications. JVM becomes an instance of JRE at run time.
- ❖ Many vendors has implemented JVM like (SUN Microsystems, IBM JRE, Oracle JRE etc.
- ❖ Sun's implementations of the JVM is itself called as JRE.
- ❖ JVM is Platform dependent.

INTRODUCTION OF JAVA

JDK (Java Development Kit)



History of java

- ❖ Java is designed by Sun Microsystems in the early 1990s to solve the problem of connecting many machines together.
- ❖ Initial version of Java is called as OAK.
- ❖ Later java became popular for developing internet applications.
- ❖ Java is developed by James Gosling and team.
- ❖ Java is publicly released on 1995.

Editions

1. Java Standard Edition (JSE)
 - Used to develop Standalone application using applet and swing.
2. Java Enterprise Edition (JEE)
 - Used to develop enterprise application using Servlets, JSP, and JDBC etc.
3. Java Micro Edition (JME)
 - Used to develop application for micro device like Mobiles, Setup Box etc.

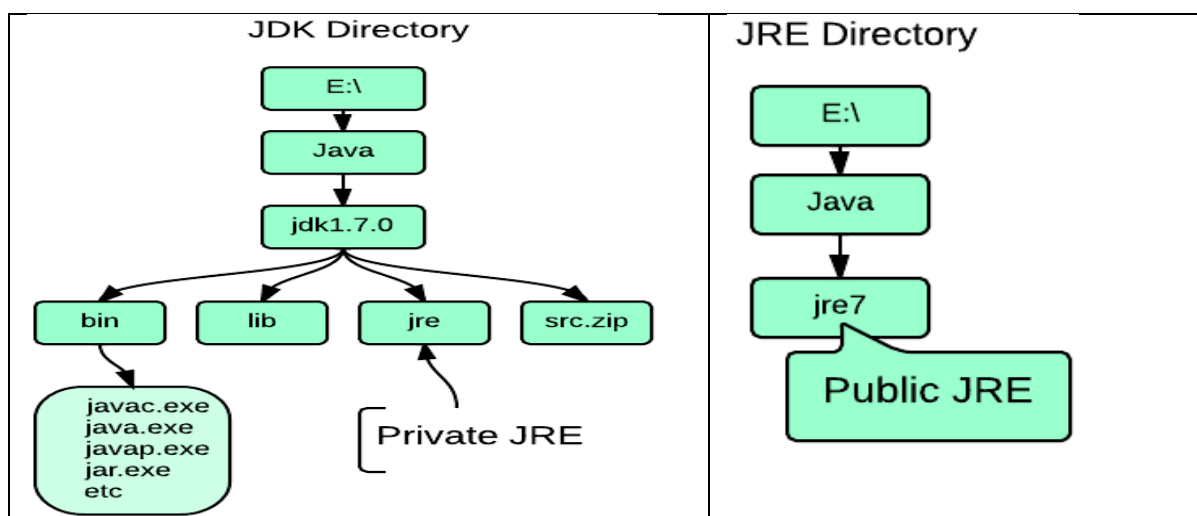
Java Standard Edition (JSE) versions

Java Versions	Code name	Released Year
JDK1.0	-	1995-96
JDK1.1	-	1997
J2SE1.2	Playground	1998
J2SE1.3	Kestrel	2000
J2SE1.4	Merlin	2002
J2SE 5.0	Tiger	2004
Java SE 6	Mustang	2006
Java SE 7	Dolphin	2011
Java SE 8	-	2014

Installation of java

- ❖ Run the installer file (like jdk 7u25-windows i586.exe).
- ❖ Click on next Button
- ❖ If you want to change the location of installation then click on change button and select that location like (E:\Java\jdk1.7.0\) → Ok → Next.
- ❖ After some time, One windows will appear, Provide the JRE location
If you want to change the location then click on change and provide the location like (E:\Java\jre7\) → Next → Close

After Installation JDK you will get the following Directory Structure.



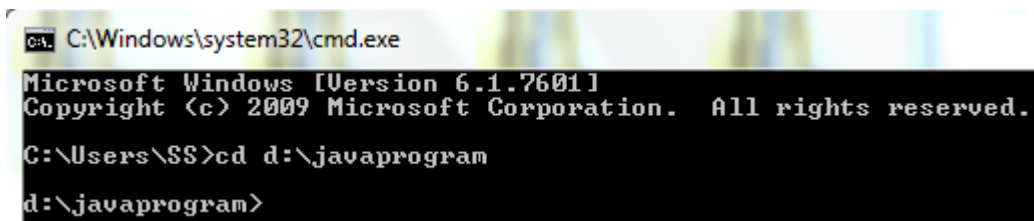
Step to Write, Compile and Run java aprogram**Write java program**

- ❖ Select the required WORKSPACE (A workspace is a directory on your hard drive (file System) where the files related to your project will be stored.
- ❖ Your selected Workspace like (D:\javaprogram\)
- ❖ Use Notepad or some other text editor to create the source file.

Step: - Open Notepad → Write a java program → Save the file with the name like Hello.java in the Directory (D:\javaprogram).

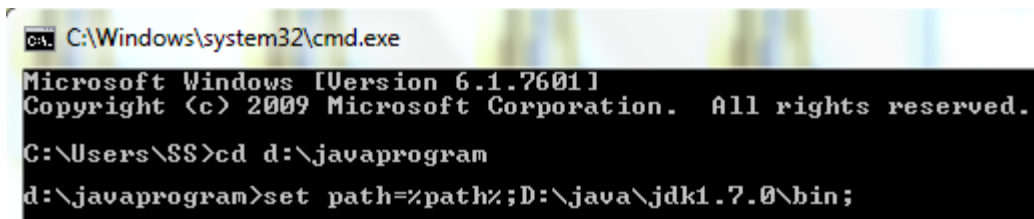
Compilation

- ❖ Click on Start → run → Type cmd → Ok
- ❖ Change the Directory to workspace directory



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.
C:\Users\SS>cd d:\javaprogram
d:\javaprogram>
```

- ❖ Set the path of bin directory of JDK.



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.
C:\Users\SS>cd d:\javaprogram
d:\javaprogram>set path=%path%;D:\java\jdk1.7.0\bin;
```

- ❖ Use javac compiler to compile the source code.
 - ⇒ Javac <Source file name>.java
 - ⇒ Example:- javac Hello.java

Run Java Program

- ❖ Use java interpreter to execute/run the java class.
 - ⇒ Java <Class file Name>
 - ⇒ Example:- java Hello

Permanently set path in System Environment Variables

Step: - Computer properties=> advanced tab=>Environment variables=>New Button

- provide the required information

Variable name	CLASSPATH
Variable value	D:\java\jdk1.7.0\bin;
- Click on ok Button.